

SoftwareVersion Field

This field SHALL be set to the value of the SoftwareVersion being downloaded, matching the [SoftwareVersion](#) field of the QueryImageResponse that caused the failing download to take place.

BytesDownloaded Field

This field SHALL be set to the number of bytes that have been downloaded during the failing transfer that caused this event to be generated.

ProgressPercent Field

This field SHALL be set to the nearest integer percent value reflecting how far within the transfer the failure occurred during the failing transfer that caused this event to be generated, unless the total length of the transfer is unknown, in which case it SHALL be null.

PlatformCode Field

This field SHOULD be set to some internal product-specific error code, closest in temporal/functional proximity to the failure that caused this event to be generated. Otherwise, it SHALL be null. This event field MAY be used for debugging purposes and no uniform definition exists related to its meaning.

11.21. Over-the-Air (OTA) Software Update File Format

11.21.1. Scope & Purpose

The majority of devices will undergo an over-the-air (OTA) software update at some point during their operational lifecycle. It cannot be assumed that the Node responsible for serving OTA updates (OTA Provider) has any specific knowledge about the internals of OTA Requestor Nodes that are receiving OTA updates. This section provides a standardized header that SHALL be included in all OTA Software Images, in order to provide the necessary information for OTA Providers to validate images available for a given OTA Requestor.

It should be noted that while this specification standardizes an OTA Software Image header that SHALL be used by all OTA Software Images, this specification does not further attempt to standardize the remaining contents of those files.

11.21.2. General Structure

The OTA Software Image file format is composed of a header followed by an opaque body. The header describes general information about the file such as software version, and the [Vendor ID](#) and [Product ID](#) for which the image applies, see [Section 11.20.3.3.2, “Conceptual algorithm for matching OTA Software Images applicable to a query”](#)).

OTA Software Image files SHALL use a fixed encoding. Individual fields of the OTA Software Image file may be comprised of more complex data types that utilize other encoding schemes.

The fields that comprise an OTA Software Image file, listed in the sequential order in which they